[PLACEHOLDER] Developement of a Segmented Communication Protocol for Sensor Data Transmission Using LoRa© Technology

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***Abstract*—The abstract goes here...**

***Index Terms*—**

I. Introduction

LOW-POWER, low-cost and long-range wireless communication technologies are becoming increasingly important in the context of the Internet of Things (IoT). Applications such as environmental monitoring, precision agriculture or Unmanned Aerial Vehicles (UAVs), require reliable data transmission over extended distances with strict power and hardware constraints.

A range of different communication standards, including ZigBee, Sigfox and Long Range (LoRa©) have been adopted to address these needs, and LoRa©, in particular, has emerged as the leading choice for low-power and long-range deployments thanks to its robust modulation technique and open network architecture. However, conventional solutions based on LoRa© (particularly those following the Long Range Wide Area Network (LoRaWAN©) specifications) are optimized for small, non-frequent payloads.

This paper presents the design and implementation of a custom communication protocol for LoRa©, engineered to enable the transmission of segmented large data payloads over long distances. The proposed system has been deployed on a sounding rocket built by the Aurora Rocketry team in the University of Bologna, demonstrating its applicability to highly dynamic systems, that require continuous transmission of large amount of telemetry data with high reliability and low power consumption in a point-to-point configuration. This approach is generalizable to a wide range of IoT applications requiring extended data capacity over extended distances and high power efficiency, and possibly to multi-agent systems that require the transmission of structured data between the agents.

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II. State of The Art

III. Proposed System

IV. Implementation

V. Testing

VI. Conclusion

The conclusion goes here.

Appendix A
Title of Appendix One

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Appendix B

Appendix two text goes here.

Acknowledgment

The authors would like to thank...

References

[1] M. Radeta, J. Pestana, P. Abreu, R. Freitas, F. Silva, D. Vieira, R. Prieto, M. Fernandez, F. Alves, T. Dellinger, S. Neves, and E. Delory, "Triton—open telemetry and location estimation for marine monitoring based on iot and lora," *IEEE Journal of Oceanic Engineering*, vol. 50, no. 2, pp. 1244–1258, 2025.

Alessandro Monticelli Biography text here.